

Such strands are schematically represented in (3). The further development of the granular membrane needs no comment in addition to the above description.

The so-called vitelline membrane of the unfertilized egg is not identical, as often assumed, with the fertilization membrane. It forms only a part of the latter. In certain circumstances, the cortical granules are not incorporated into the membrane. Under these conditions, a thin membrane results which has a low double refraction and may be identical with the vitelline membrane.

The description given above of the fate of the cortical granules is at variance with the ideas entertained by Moser² concerning their role in the activation.

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¹ Runnström, J., Monné, L., and Broman, L., *Arkiv f. Zoologi* (Stockholm), **35**, A, Nr. 3 (1943).

² Runnström, J., *Protoplasma*, **4**, 388 (1928).

³ Hendee, Ester C., Papers from Tortugas Lab., Carnegie Inst. Pub., **27**, 101 (1931).

⁴ Lindahl, P. E., *Protoplasma*, **16**, 378 (1932).

⁵ Moser, F., *J. Exp. Zool.*, **80**, 423 and 442 (1939).

⁶ Cf. Runnström, Tiselius and Lindvall, *NATURE*, **153**, 285 (1944).

⁷ Just, E. E., *Biol. Bull.*, **36**, 1 (1939).

Water Contents of Last-stage Larvæ, Pupæ, and Adults of the Meal Moth

ACCORDING to Speicher¹, larvæ and pupæ of the meal moth, *Ephestia kuehniella* Zeller, an important pest in meal stores, maintain a constant percentage of free water, independent of the relative humidity of the environment.

In contrast to Speicher's results, our observations show that migrating larvæ of *Ephestia*, which have been reared at low relative humidities, have a much lower average water content than larvæ kept at high humidities, although the water contents show considerable fluctuation.

On the other hand, water contents of pupæ of all ages show a practically steady average figure, which is independent of the relative humidity in which the insects were reared. This figure (see accompanying table), which agrees very well with Speicher's value, lies between the average percentage water contents of larvæ reared at 30 per cent and 70 per cent relative humidity respectively.

	Average water content as % of body weight	
	30% relative humidity	70% relative humidity
Migrating larvæ	57.3	73.5
Pupæ	65.4	66.4
Adults	65.4	66.3

The average water content of adults is practically the same as that of pupæ, and again appears to be independent of the humidity of the environment.

It is generally held that the water content in various instars of insect development can be regulated by the retention of metabolic water². It seems, however, that *Ephestia* larvæ cannot compensate

entirely for the increased rate of evaporation of water at low relative humidities.

When larvæ reared at low relative humidities are preparing to pupate, a more efficient mechanism of regulation comes into operation, so that the water contents of the resulting pupæ are almost identical with those reared at high humidities. The average weight of these pupæ developing at low humidities (30 per cent relative humidity), however, is approximately only 80 per cent of those reared under standard conditions (70 per cent relative humidity). It may be that the regulatory process involves the oxidation of a greater weight of reserve substances during the pre-pupal stage of those individuals reared in the drier environment.

Low environmental humidity definitely increases the duration of larval development (fifty days at 30 per cent relative humidity, as compared with thirty-three days at 70 per cent relative humidity), but the length of the pupal instar is only slightly affected.

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¹ *Proc. Pennsylvania Acad. Sci.*, **5**, 79 (1931).

² Wigglesworth, V. B., "Principles of Insect Physiology", 354 (London, 1939).

New Interference Phenomena with Newton's Rings

SOME striking new interference phenomena have been found with Newton's rings by using multiple-beam interference instead of the usual two beams. The rings, formed between a convex lens and a flat piece of glass, are modified profoundly by the employment of multiple beams. The two surfaces in contact are coated (by evaporation *in vacuo*) with high reflecting coefficient transparent silverings, the reflecting coefficient exceeding 0.85 for the green mercury line. The resulting multiple beams lead to the production of fringes which are characterized by their remarkable sharpness. The fringes in transmission are fine narrow brilliant rings on a broad dark background, and in reflexion can be seen a complementary system of fine dark 'absorption' fringes on a broad bright background.

Normal Incidence. Typical transmission rings (green mercury) with light incident normally on the interference faces are shown in Fig. 1. The sharpness is unique. It is necessary to restrict the incident light to a single angle of incidence by employing a small source at the focus of a lens in order to achieve the best definition. Comparison with the ordinary classical two-beam rings reveals the superiority of the multiple-beam fringes in the following particulars:

(1) They are much more intense.
(2) They are so sharp that very high precision can be attained. A change of 1/100 of an order can readily be accurately measured. This corresponds to a displacement of only 25 angstroms between the optical components. A great increase in precision is thus now available in all the numerous metrological applications of Newton's rings.

(3) The fringes are so inherently sharp that fine-scale surface defects and irregularities on the glass surfaces are rendered as detail in the fringes. For the reproduction shown as Fig. 1, the flat component was a simple piece of glass, and all the local

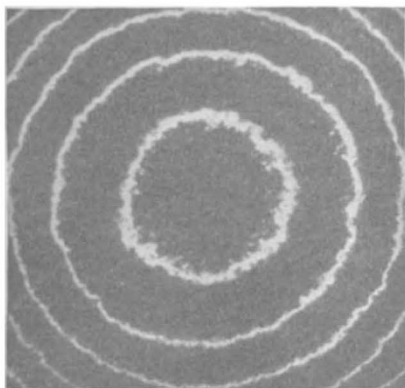


Fig. 1. NEWTON'S RINGS WITH MULTIPLE BEAM INTERFERENCE.

irregularities are clearly shown by the irregularities of fringes.

(4) By matching the curved component against a high-grade optical flat, the perfection of figure of the lens surface is revealed, with very great sensitivity. This effectively extends to curved surfaces the existing multiple-beam Fizeau test, used to-day only for flat surfaces.

Non-normal Incidence. New interference phenomena appear when the angle of incidence is other than normal. The fringes are no longer located in the film but lie on focal curves, one half of the ring system being situated before and the other half behind the interference film.

The fringes become double; the separation between the two components growing steadily with increasing incidence. One component moves *out* towards the next higher order, becoming progressively *weaker* and also *sharper*. These two components are plane polarized, mutually perpendicularly. The phenomena are shown clearly by Fig. 2, which refers to an incidence of 60° . Fig. 2b shows the doubling, with no Nicol prism in the beam. In Fig. 2a a Nicol has been interposed to pass the beam with the magnetic vector perpendicular to the plane of incidence; and in Fig. 2c the Nicol has been set to pass the parallel component. Owing to the peculiar focal

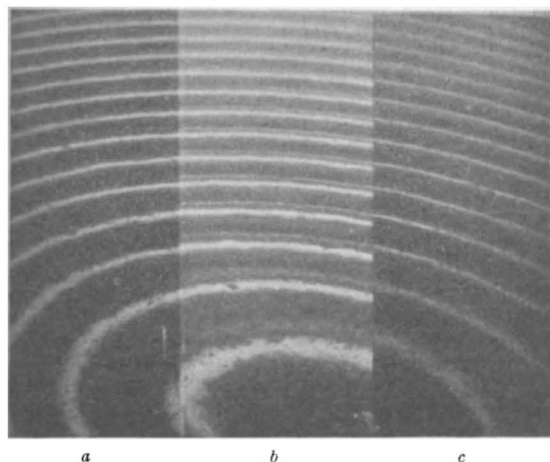


Fig. 2. DOUBLING DUE TO DIFFERENTIAL PHASE CHANGE AT REFLEXION. (a) MAGNETIC VECTOR PERPENDICULAR TO PLANE OF INCIDENCE; (b) NO NICOL PRISM; (c) MAGNETIC VECTOR PARALLEL TO PLANE OF INCIDENCE.

properties of the non-normal fringes, only the fifth and sixth fringes are in focus in this reproduction.

The cause of the doubling is the existence of a differential phase change at reflexion at the silver surfaces. The phase change is different for the mutually perpendicular directions of vibration of the light. This differential effect has been measured over the range of incidence $0-80^\circ$. To the first order, the observations closely confirm the predictions of classical electromagnetic theory. Up to an incidence of 60° the experimental curve runs closely parallel to the theoretical curve, but is displaced upwards by a quite small, but real, amount, 0.007λ . The observations reveal a marked, but small, inflexion at 63° , not predicted by theory, which is probably incomplete as to finer detail.

The remarkable sharpness of the parallel vector (see fifth and sixth fringes) is due to the higher effective Fresnel reflecting coefficient for this vector. It can be demonstrated that this parallel vector is rapidly absorbed with increasing incidence, and the rate has been determined. It follows from this that an evaporated silver film, when used to transmit light incident at high angles of incidence, should act as a polarizer (partial) in a manner similar to that which makes tourmaline act as a polarizer by differential absorption. This interesting property has been shown to exist, and can be very easily demonstrated.

The same characteristic fringe-doubling due to the differential phase change at reflexion at the silver has been found with small gap Fabry-Perot interferometers, with the necessary modification that the parallel vector moves *inwards*, to the region of higher order of interference. For gaps of less than 1 mm., the differential doubling can lead to an appreciable loss in resolution, if the incident light is not polarized to compensate for this.

Full experimental details, together with an analysis of the phenomena very briefly surveyed here, have been communicated elsewhere.

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Influence of an Adsorbed (Inner) Layer on the Cohesion of a Solid

IN NATURE of January 22, Benedicks and Sederholm, under the above title, showed that the exposure to air of freshly cut surfaces of paraffin, lead and tin resulted in a lessened cohesion when two plane surfaces of the solid in question are brought into contact. By measuring the cohesion after different periods of exposure to the air, it was shown that, in the case of tin for example, important changes took place during the first five minutes exposure of a freshly cut surface. These changes are attributed to the rapid formation of a thin adsorbed layer.

A similar effect has been observed with silver and with iron by another simple technique. When a drop of distilled water of volume 0.03 c.c. is allowed to fall a distance of 2 cm. from the jet of a burette on to a freshly cut surface of the metal fixed in a horizontal position, the diameter to which the drop spreads before coming to rest is governed to a certain extent by the length of time the metal surface has been exposed to air.